

1. Compare the hashing and Binary Search Based on time Complexity and methods.
2. Compare the time Complexity of all notations in Bubble, insertion, selection, merge, heap, Quick sort

CO4 - A71

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1. Hashing :-

Hashing is a technique used to store and retrieve data using hash function.

2. Binary Search :-

Binary Search is a searching technique used to sort an array.

Time Complexity

	Hashing	Binary Search
Best	$O(1)$	$O(1)$
Average	$O(1)$	$O(\log n)$
worst	$O(n)$	$O(\log n)$

Basic method

Hash-function +
hash table

Divide the
sorted array
into half

Searching

very fast on
Average

slower than
hashing.

issue

Collision

Requires sorted
Data.

Conclusion:

hashing provides $O(1)$ average search time, making it very efficient for direct key-based searching.

Binary Search provides $O(\log n)$ searching but requires the data to be sorted.

Algorithm

Bubble
sort

insertion sort

selection sort

Merge sort

Heap sort

Quick sort

Best Case

$O(n)$

$O(n)$

$O(n^2)$

$O(n \log n)$

$O(n \log n)$

$O(n \log n)$

worstCase

$O(n^2)$

$O(n^2)$

$O(n^2)$

$O(n \log n)$

$O(n \log n)$

$O(n \log n)$

Average Case

$O(n^2)$

$O(n^2)$

$O(n^2)$

$O(n \log n)$

$O(n \log n)$

$O(n^2)$